Comment on "Entropic uncertainty principle for mixed states": An exact qutrit counterexample to the extended-MUB conjecture

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Rotundo and Schwonnek conjectured that a condition involving only the second singular value of a doubly stochastic measurement-overlap matrix makes the uniform vector a global maximizer of a related induced norm. We give an exact counterexample within the physically realizable qutrit subset. Two complete rank-one projective measurements are related by a rational real orthogonal matrix, so their overlap is orthostochastic and hence unistochastic. At $\lambda = 3/4$ and $\mu = 1/3$, the proposed singular-value condition holds with a strict exact margin, but the positive vector $(6, 10, 5)^{\mathsf{T}}$ produces a norm ratio strictly larger than the mutually unbiased-basis value. The two strict inequalities reduce to a quadratic-radical comparison and a positive integer certificate. Thus local stability at the uniform vector does not imply the conjectured global formula, even for two legal qutrit measurements. The proved induced-norm uncertainty relation itself is unaffected.

I. CONTEXT AND CLAIM

For two complete rank-one projective measurements X and Y in dimension d, let

$$C_{ij} = \mathrm{Tr}(X_i Y_j) = |U_{ij}|^2, \qquad U \in U(d), \tag{1}$$

where the columns of U express the Y basis in the X basis. Rotundo and Schwonnek derived, at entropy weight $\alpha = 1$, the uncertainty relation

$$\lambda H_X(\rho) + \mu H_Y(\rho) \geq S(\rho) - \log \|C\|_{r\to s}, \tag{2}$$

where $r = 1/\mu$ and $s = 1/(1-\lambda)$, and proposed an analytic formula for the norm in an "extended MUB regime" [1]. In the nontrivial domain $0 < \lambda, \mu < 1$ and $\lambda + \mu > 1$, their conjecture states that

$$\frac{(1-\mu)(1-\lambda)}{\mu\lambda} \geq \sigma_2(C)^2 \quad \Longrightarrow \quad \|C\|_{r\to s} = d^{\,1-\lambda-\mu}. \tag{3}$$

Here $\sigma_2(C)$ is the second singular value. Because C is entrywise nonnegative, its induced norm may be written as

$$\|C\|_{r\to s} = \max_{x\in\mathbb{R}_+^d\setminus\{0\}} \frac{\|Cx\|_s}{\|x\|_r}. \tag{4}$$

The uniform vector always gives the right-hand side of Eq. (3) as a lower bound. The supplemental analysis of Ref. [1] shows that the singular-value condition makes this vector a local maximum; the missing step is global optimality.

We disprove Eq. (3) already for $d = 3$, while retaining the physical restriction $C_{ij} = |U_{ij}|^2$. No relaxation to an arbitrary doubly stochastic matrix is used.

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II. EXACT PHYSICAL COUNTEREXAMPLE

Take the real matrix

$$U = \begin{pmatrix} 1/3 & -14/15 & 2/15 \\ -2/3 & -1/3 & -2/3 \\ 2/3 & 2/15 & -11/15 \end{pmatrix}. \tag{5}$$

Direct rational multiplication gives $U^{\mathsf{T}}U = I_3$ and $\det U = 1$. Hence U is an exact unitary matrix on $\mathbb{C}^3$, and its columns define a complete rank-one projective measurement. Equation (1) gives

$$C = |U|^2 = \begin{pmatrix} 1/9 & 196/225 & 4/225 \\ 4/9 & 1/9 & 4/9 \\ 4/9 & 4/225 & 121/225 \end{pmatrix}. \tag{6}$$

Every row and column sums to one. More importantly, Eq. (5) certifies that C is orthostochastic, and therefore unistochastic.

Choose

$$\lambda = \frac{3}{4}, \qquad \mu = \frac{1}{3}, \qquad r = 3, \qquad s = 4. \tag{7}$$

The parameter conditions are satisfied because $\lambda + \mu = 13/12 > 1$. To check spectral admissibility exactly, the characteristic polynomial of $C^{\mathsf{T}}C$ factors as

$$\det(tI - C^{\mathsf{T}}C) = (t-1)\frac{50625t^2 - 33714t + 49}{50625}. \tag{8}$$

Its nontrivial eigenvalues are

$$\eta_\pm = \frac{1873 \pm 64\sqrt{849}}{5625}, \qquad \sigma_2(C)^2 = \eta_+. \tag{9}$$

For Eq. (7), the threshold in Eq. (3) equals $2/3$, and

$$\frac{2}{3} - \sigma_2(C)^2 = \frac{1877 - 64\sqrt{849}}{5625} > 0. \tag{10}$$

The last inequality is exact: both terms being compared are positive and

$$1877^2 - 64^2\,849 = 45625 > 0. \tag{11}$$

Thus the proposed condition holds strictly, so the uniform vector is locally stable under the cited second-variation test.

It is not a global maximizer. Set $x = (6, 10, 5)^{\mathsf{T}}$, which lies in the interior of the positive cone. Then

$$Cx = \left(\frac{142}{15}, 6, \frac{83}{15}\right)^{\mathsf{T}}. \tag{12}$$

Define

$$A = \sum_i (Cx)_i^4 = \frac{519655217}{15^4}, \qquad B = \sum_i x_i^3 = 1341. \tag{13}$$

For $R = \|Cx\|_4/\|x\|_3$, one has $R^{12} = A^3/B^4$. The conjectured MUB value is $3^{-1/12}$, whose twelfth power is $1/3$. Their strict ordering is certified entirely in integer arithmetic:

$$\begin{aligned} &3(519655217)^3 - 15^{12}(1341)^4 \\ &\qquad = 14098963811343952951693 14 > 0. \end{aligned} \tag{14}$$

Consequently,

$$\|C\|_{3\to 4} \geq R > 3^{-1/12} = 3^{\,1-\lambda-\mu}. \tag{15}$$

One feasible vector is sufficient for this strict lower bound; determining the global value of $\|C\|_{3\to 4}$ is unnecessary for the disproof.

III. CONSEQUENCES AND LIMITATIONS

Equations (10) and (15) separate the local and global questions sharply: the uniform vector passes the strict singular-value stability test, yet another positive vector gives a larger objective value. Thus $\sigma_2(C)$ alone cannot certify the MUB-form norm replacement, even for two complete rank-one qutrit measurements connected by an exact real unitary.

This conclusion concerns only the conjectured evaluation of the induced norm. It does not alter the proved bound in Eq. (2), which remains valid with the actual norm. Nor does the counterexample by itself exhibit a state that violates an entropic uncertainty inequality. It shows instead that using the MUB value at this admissible point would require a stronger sufficient criterion or a separate norm computation. The example also does not characterize the full parameter region in which the MUB formula may remain valid.

DATA AND CODE AVAILABILITY

All quantities needed for the exact argument are displayed above. A machine-readable certificate and a deterministic verification script using only Python's standard library accompany the manuscript. The script checks orthogonality, stochasticity, spectral admissibility on the sum-zero subspace, and the positive integer norm certificate without floating-point arithmetic or network access.

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OpenAI Codex with the gpt-5.6-sol model and Ultra reasoning setting was used substantively for literature triage, exact-arithmetic code generation, verification design, and manuscript drafting under prompts that fixed the physical domain and required independent exact checks. The reported arithmetic was cross-checked by structurally different exact-rational implementations. Human author review of the science, literature, authorship details, and wording remains pending; this local draft is not authorized for submission.

[1] A. F. Rotundo and R. Schwonnek, Entropic uncertainty principle for mixed states, Physical Review Research **6**, 033043 (2024).